

What drives predictive accuracy and response times in incremental sequence learning?

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how do language users learn linguistic sequences?

relevance of sequence learning to linguistics

sequence learning is necessary for **almost every level and modality of language**, e.g.:

- most famously, **word segmentation** (Saffran, Newport, & Aslin, 1996)
- but also **speech comprehension** – co-articulation, learning words, morphology, syntax, etc.
- **speech and language production** – incl. speech, signing, writing, typing...

research question

We investigated **which learning mechanism underlies incremental learning of sequences**

- transitional probability or error-driven learning?

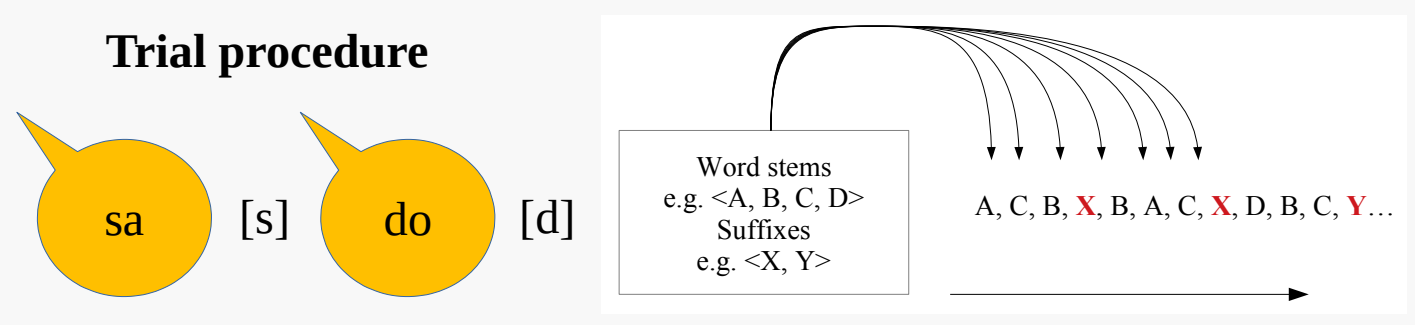
Exp 1: reaction times identifying spoken syllable, e.g. hear ‘sa’, press [s] key (Tomaschek et al., 2024)

Exp 2: same stimuli as Saffran et al. (1996), but incrementally predicting **upcoming syllable**

Experiment 1: Reaction time identifying spoken syllable

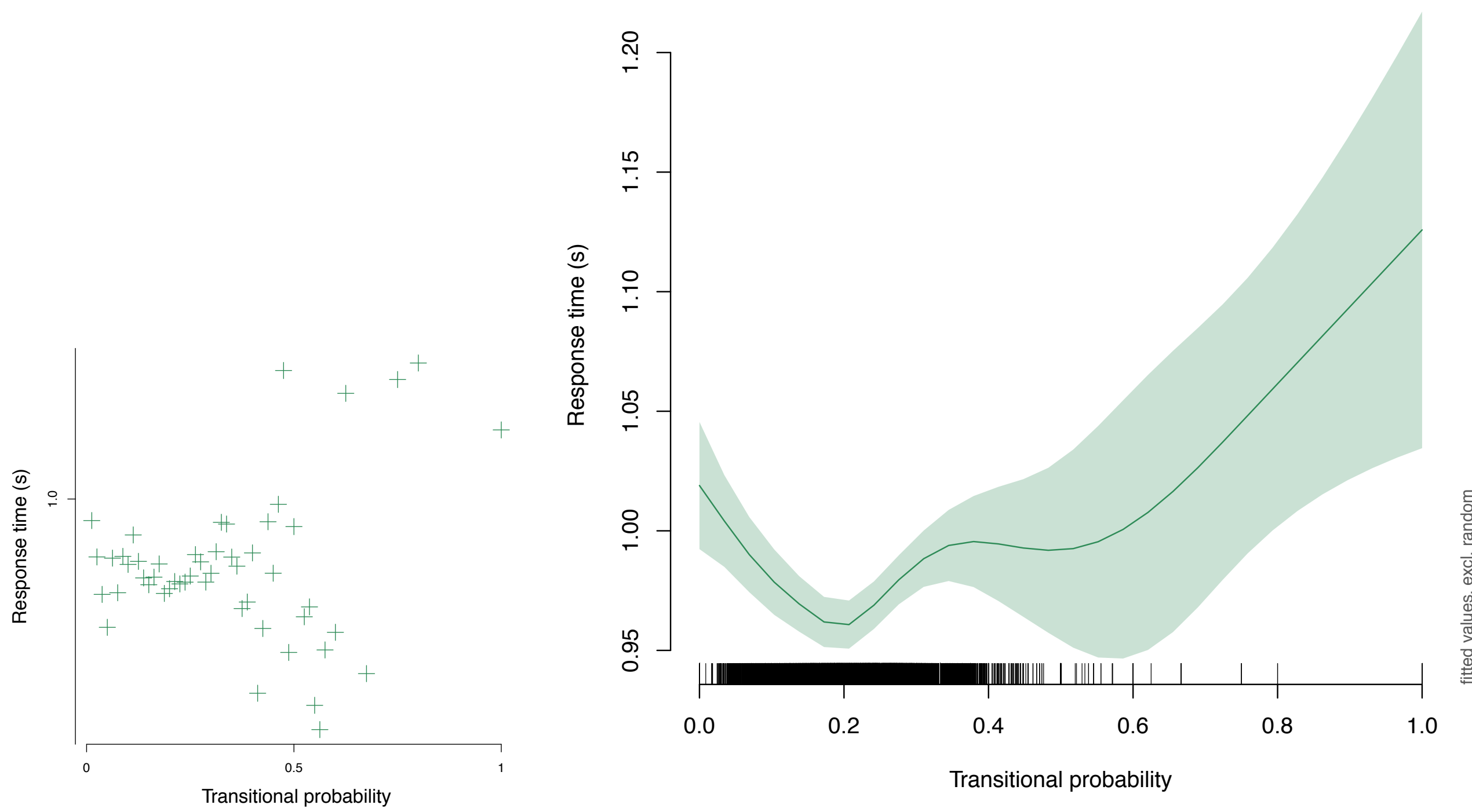
method Experiment 1

- 16 native German participants
- heard syllable, typed corresponding key
- syllables: sa, do, fo, je, ke, la
- keys: s, d, f, j, k, l

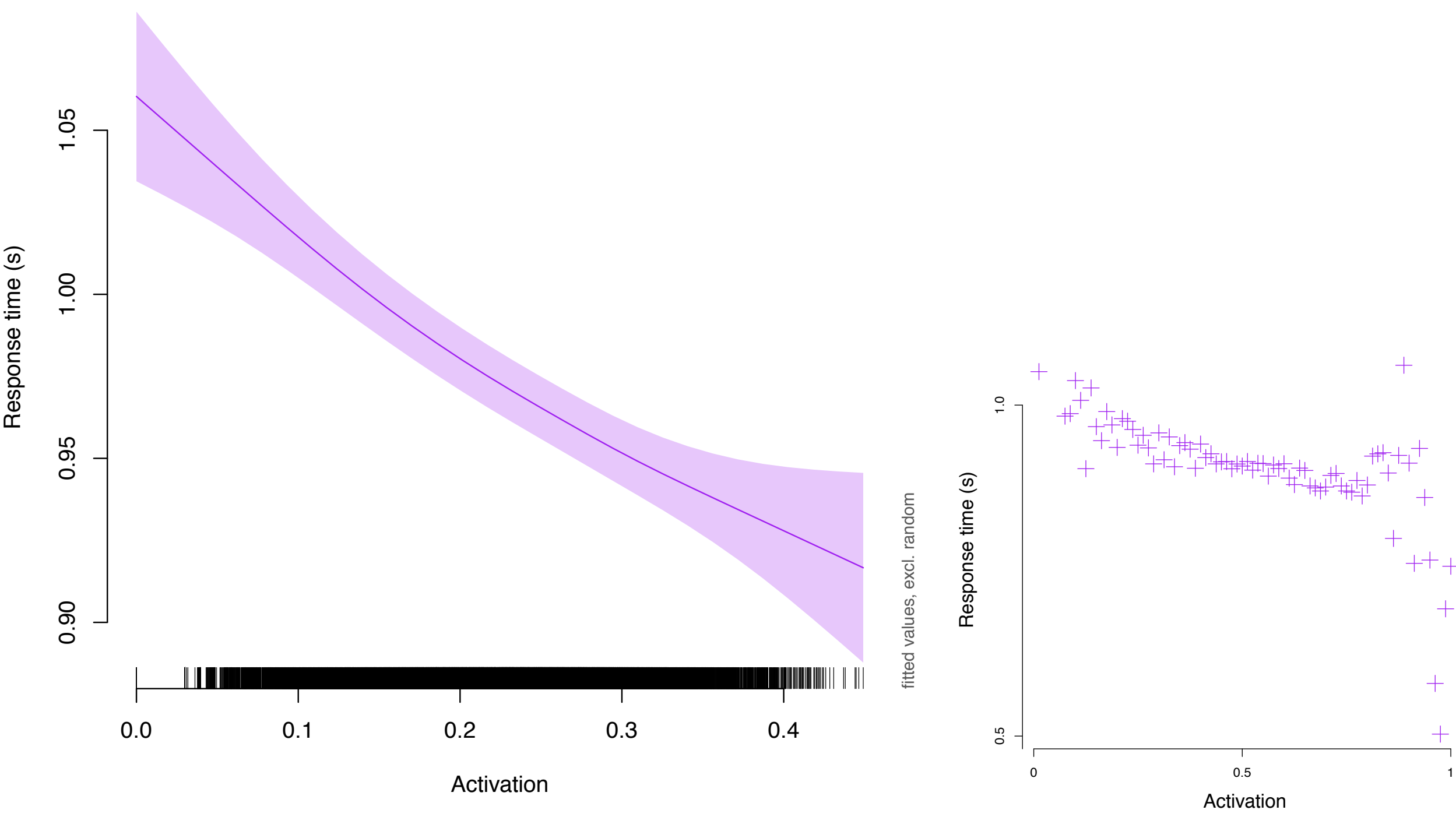


results Experiment 1: simulation visualisation (raw data) and GAMM plots

Effects of transitional probability on RT



Effects of error-driven learning (activation) on RT



Experiment 2: Accuracy predicting upcoming syllable

method Exp 2

Based on Saffran et al. (1996), running speech stream ...

But to investigate **incremental learning and prediction**

→ speech stream stopped at pseudorandom intervals (marked with [prompt])

**babupututibubupa[prompt 1]...dababupubupadadu[prompt 2]...
tabapatubipidabututibudutaba[prompt 3]...babupubupadapida
[prompt 4]...bupatubitutibubabupu[prompt 5]...bupadatutibu
patu[prompt 6]... bidutaba.....**(and so on; 6 words, 300 x each; no prosodic cues)

16 native German participants were asked to **type the next syllable...**

Correct responses: da... ta... ba... bu... bu... bi...

simulations: two different learning measures

for each participant on each trial

transitional probability

babupubupadadutabababupututi**bu**bu

da...
babupubupadadutabababupututi**bu**pa

da...
babupubupadadutabababupututi**bu**pa**da**...

transitional probability of **da** given **pa** =

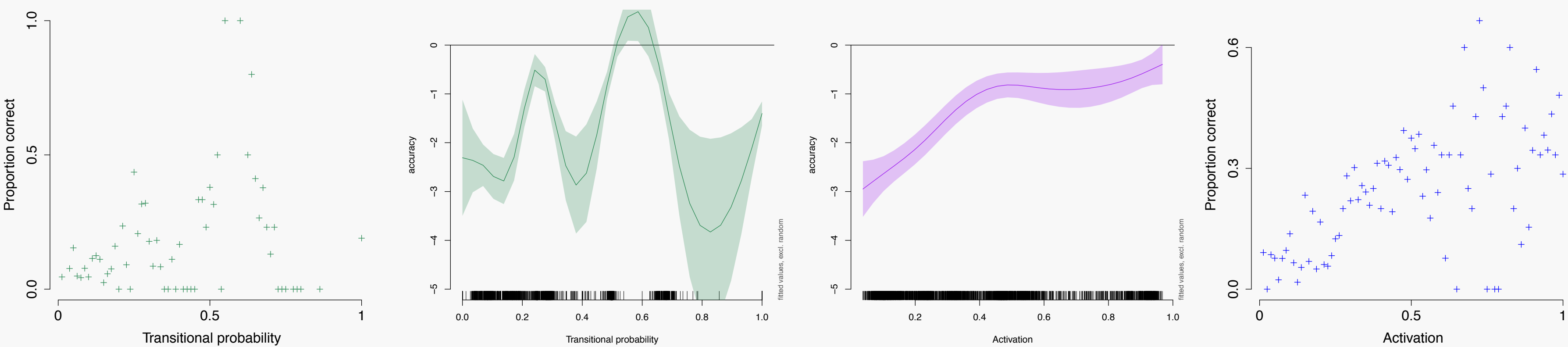
frequency of pada

frequency of pa

error-driven learning: ‘activation’

babupubupadadutaba... : #ba ba# → **bu**
ba**bu**pupadadutababa : #bu bu# → **pu**
babu**pu**bupadadutababa : #pu pu# → **bu**

Rescorla-Wagner learning equations:(1)
develop expectations about next syllable from
cue sequence



background

Saffran et al. showed listeners can **learn from structure in sequences**

- participants selected ‘words’ more often than chance in a forced-choice task after training
- interpreted as learning from **transitional probability**

Since then, much work *assumes* sequence learning involves transitional probability

but our recent work suggests phonetic learning may instead be error-driven:

- Phonetic learning is error-driven, not just statistical (Nixon, 2020, *Cognition*)
- error-driven learning model of infant learning from acoustic signal (Nixon & Tomaschek, 2020, 2021, *Cognition*)
- Response times in sequence learning predicted by error-driven learning (Tomaschek, Ramscar, & Nixon, 2024, *Cognitive Science*)

Error-driven learning: a predictive process that discriminates between informative and uninformative cues
(Nixon, 2020; Ramscar, Dye, & McCauley, 2013)

discussion and conclusion

- Saffran et al used a 2AFC task *after training* using summed transitional probabilities
- the present study tests *incremental learning*:
 - simulated degree of expectation of each current (Exp 1) or upcoming syllable (Exp 2) *throughout the experiment*
 - EDL improved predictions of by-trial RT (Exp 1) and accuracy (Exp 2), relative to TP

Transitional probability may be a good approximation on **summed data** (e.g. Saffran et al., 1996)

→ but with the increased resolution of **trial-by-trial testing** we see that **error-driven learning** seems to be the better model of incremental sequence learning